

Polymerase Chain Reaction (PCR)

Preparation

Material ※	Concentration	Addition per sample	Location
gDNA [†]	10 ng / uL	4 µL	Usually -20°C
F-primer		0.4 µL	-20°C (2row-3col-4layer)
R-primer		0.4 µL	-20°C (2row-3col-4layer)
2X-Taq		5 µL	-20°C (4row-3col-4layer; orange lid)
Ambion water		0.2 µL	RNA bench (right-hand side, top drawer)
Total :		10 µL	

※ All materials should be placed on ice.

[†] gDNA : The gDNA stock solution of all samples should be standardized to the same concentration, for example 10 ng/µL.

Procedure

1. Prepare the total amount of the required working solution (F-primer, R-primer, 2X-Taq);

- For example, if we had 10 samples, then we make the working solution by adding
4 μ L **F-primer**,
4 μ L **R-primer**,
and 50 μ L **2X-Taq** into a new 200 μ L PCR tube.

2. Dispense the working solution into a new 200 μ L PCR tubes (5.8 μ L for each sample);

3. Lastly, add the sample gDNA stock solution;

4. Close the lid tightly, and swipe the tube gently with finger several times to mix the solution well;

5. Slightly centrifuge to bring down the solution;

6. Put into the PCR machine (the machine should be warmed-up around 2 mins before use).

Reaction curve:

Process	Temperature (°C)	Time	Cycle
Initial denaturation	95	1 min	--
Denaturation	95	15 secs	30
Annealing [†]	56	15 secs	30
Extension	72	1 min	30
Complete extension	72	5 min	--
PCR product storage	4	∞	--

[†] Annealing temperature depends on primers.

Done !!!
